

HITIK PANCHAL

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SUMMARY

Computer Science Master's student at Universität Bonn (Grade: 1.7) with 1.5 years of production experience building high-throughput backend systems at NSE India. Hands-on professional experience with Python, Java/Spring Boot, SQL, and REST APIs across large-scale financial systems. University research focused on NLP and LLMs - including LoRA fine-tuning of transformer models and multilingual narrative classification using RoBERTa and LLaMA. Personal projects spanning full-stack websites, agile planners and tableau dashboards.

PROFESSIONAL EXPERIENCE

Associate System Analyst · NSEIT Limited (now DEXIT Global)

May 2023 – Feb 2024 · Mumbai, IN

- **Project lead** for NCMS CM module - owned delivery planning, stakeholder communication, and release coordination across a cross-functional team of engineers and QA analysts. Mentored 2 junior developers through structured code reviews, standardizing review criteria to catch issues earlier in the cycle.
- Architected and delivered migration of NSE Clearing Management System (NCMS) from **Java/Spring Boot monolith to REST API-based architecture**, increasing throughput by **~20%** and enabling the platform to sustain **20,000+ trades/minute** for live clearing operations.
- Identified and resolved critical backend performance bottlenecks; DB query refactoring reduced database load by **~10%** and eliminated several production bugs, improving overall system stability by **4-5%**.

Stack: Java, Spring Boot, MySQL, JavaScript, Linux, REST APIs, Postman, Git

Trainee Associate System Analyst · NSEIT Limited (now DEXIT Global)

Nov 2022 – May 2023 · Mumbai, IN

- Executed production deployments on Linux for NCMS serving **100,000+ end users**, maintaining near-zero downtime across all release cycles.
- **Led Git migration** of **10+** legacy NCMS codebases, standardising version control across the team and significantly reducing branch conflicts and manual merge overhead.
- Resolved critical **SQL** and application bugs during **UAT** and testing phases, reducing defect backlog and accelerating time-to-production for key features.

Stack: Java, Spring Boot, SQL, JavaScript, Linux, Git

Data Science Intern · Dezinolics Web and Software Company

Aug 2022 – Oct 2022 · Navi Mumbai, IN

- Built an end-to-end customer churn prediction pipeline (logistic regression + decision tree ensemble, scikit-learn) on **200,000+ subscription records**; optimised for AUC score and delivered risk scores surfaced via Tableau dashboards adopted by the marketing team for retention targeting.
- Engineered **ETL workflows** for data cleaning, feature engineering, and model tuning, streamlining the path from raw data to production-ready model inputs.

Stack: Python, scikit-learn, Tableau, SQL, ETL pipelines

EDUCATION

M.Sc. Computer Science · Universität Bonn

Apr 2024 – Present · Bonn, DE

Current Grade: **1.7** · Focus: Machine Learning, NLP, Large Language Models

B.E. Computer Engineering · Mumbai University

Jun 2018 – Mar 2022 · Navi Mumbai, IN

CGPA: **9.76 / 10** · Final-year thesis on real-time crowd density estimation using CNN/MCNN (~90% accuracy)

PERSONAL PROJECTS

26/27 Football Season Simulator *React · TypeScript · Zustand · Web Workers · Tailwind*

- Built a **live** browser-based **football management simulator** (26-27-football-season-simulator.vercel.app) modeling transfers, squad building, and full 38-match seasons for 12 European clubs (~3,500 real players) using a Dixon-Coles + Monte Carlo match engine and a full domestic/European cup engine, all running client-side on seeded PRNGs.

Got You CoVered - AI Job-Application Suite *Next.js · TypeScript · Anthropic Claude API · Tailwind · Vercel*

- Built and deployed a **live** full-stack **job-application platform** (got-you-covered-cv.vercel.app) that generates **tailored cover letters, scores and optimizes your CV** against any job description (editing your original .docx in place, layout preserved), and **tracks every application** with live stats - supporting one-click .docx/.pdf export, with all user data kept client-side.

Value Alignment via Domain-Specific LoRA Fine-Tuning *Python · PyTorch · Qwen3 · LoRA/PEFT · Hugging Face*

- Researched the effect of domain-specific pretraining on value orientations of Qwen3 models (0.6B-8B) using **LoRA fine-tuning** on five subsets of The Pile; evaluated value shifts against the Kiesel Value Survey (Schwartz theory). Found that shifts in smaller models were driven primarily by output homogenisation rather than substantive value acquisition - a finding with direct implications for responsible LLM deployment.

Multilingual Narrative Classification in News Articles *Python · PyTorch · RoBERTa · LLaMA-3.1-8B · Hugging Face*

- Built a propaganda and narrative detection system for English/Russian news using a dual-head **RoBERTa** multi-label classifier with **LLaMA-3.1-8B-Instruct** refinement for low-resource classes. Achieved **85% multi-label accuracy** on a highly imbalanced dataset - approximately **18pp above baseline** - enabling automated monitoring of disinformation narratives at scale.

Sprint Copilot - AI-Assisted Agile Planning *Python · Plotly · NetworkX · Docker · Azure Container Apps · GitHub Actions*

- Shipped a live Streamlit dashboard where an AI copilot answers agile-planning questions (capacity analysis, blockers, delivery risk, forecasts) via Claude tool-use. Ingests demo data, Jira/Linear CSV exports, or a live read-only Jira Cloud connection.
- Containerized the app with Docker and deployed it to Azure Container Apps via a GitHub Actions CI/CD pipeline; [live deployment](#).

Financial Portfolio Analysis Dashboard *Tableau · Python · Yahoo Finance API · Excel*

- Designed a Tableau dashboard comparing portfolio performance across equities, bonds, and ETFs using historical Yahoo Finance data; integrated dynamic filters and comparative charts enabling side-by-side risk/return analysis - demonstrating applied financial analytics and BI skills directly relevant to investment or fintech contexts.

TECHNICAL SKILLS

Languages	Python, Java, JavaScript, C/C++, Shell, SQL, R
ML & Data Science	PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, RoBERTa, Embeddings, Pandas, NumPy
AI Engineering & Gen AI	Generative AI, RAG Pipelines, LLaMA, Qwen3, LoRA (PEFT), LangChain, Prompt Engineering, Claude Code, Google Gemini, OpenAI Codex, AI Agents (CrewAI, AutoGPT, LangGraph), MCP
Data & Viz	Tableau, Power BI, Google Data Studio, Streamlit, Excel
Backend & APIs	Spring Boot, REST APIs, MySQL, MongoDB, Firebase, Docker, Kubernetes (familiar), Azure Container Apps, GitHub Actions (CI/CD), Azure Cloud
Tools, DevOps & Automation	Git, Jira, Linux, Postman, Confluence, LaTeX, Microsoft Office, Zapier, n8n, Make (Workflow Pipelines)
Agile / PM Analytics	Sprint Planning, KPI Tracking, Capacity Analysis, Backlog Refinement, Dependency Tracking

CERTIFICATIONS

Google Data Analytics Professional Certificate (Coursera) · IBM Data Science Professional Certificate (Coursera)
Google Business Intelligence Professional Certificate (Coursera) · Data Visualization with Tableau (UC Davis, Coursera)

ADDITIONAL INFORMATION

Languages: English (Fluent, IELTS 7.5) · Hindi (Fluent) · German (A2, actively improving)

Interests: Football analytics · Chess · Photography · Travelling · Gaming